Quiz Questions

Q1: Programming, Data Structure and Algorithms (1)
If the height of the binary tree is 7 edges, maximum number of internal nodes the tree can have is,
A. 128
B. 127
C. 64
D. 63
Correct Answer: B. 127

Q2: Programming, Data Structure and Algorithms (10)
The following is the result after first partition of quicksort algorithm 10,8, 6, 9, 11, 14, 16, 15. Which of the following is true?
A. 11 could be the pivot , 14 could not be pivot.
B. 14 could be the pivot , 11 could not be pivot.
C. Either 11 or 14 could be pivot element
D. 11 and 14 both are pivot elements.
Correct Answer: C. Either 11 or 14 could be pivot element

Q3: Programming, Data Structure and Algorithms (100)
What is the potential advantage of recursive approach than an iterative approach?
A. Consumes less memory
B. Less code and easy to implement
C. Consumes more memory
D. All the options
Correct Answer: B. Less code and easy to implement

Q4: Programming, Data Structure and Algorithms (101)
Consider the array LA = {1, 2, 4, 3} is sorted using bubble sort. How many iterations is required in order to sort the array?
A. 4
B. 2
C. 1
D. 0
Correct Answer: B. 2

Q5: Programming, Data Structure and Algorithms (102)
Consider the following code snippet
```
 char c[]=”VITEEE2022”;
 char *p=c;
 printf(“%s”,p+p[1]-p[3]);
```
A. EE2022

B. EEE2022
C. E2022
D. Error
Correct Answer: A. EE2022

Q6: Programming, Data Structure and Algorithms (103)

```
int main()
 {
        int s[5];
        //considering the initial memory location of s is 1000 and one integer is of 4 byte
        s++;
        printf("the array is at memory location %d",s);
        return 0;
}
```

Predict the output
A. 1002
B. 1004
C. 1020
D. Error
Correct Answer: D. Error

Q7: Programming, Data Structure and Algorithms (11)
Run time of the recurrence T(n) = 7T(n/3)+n^2 log n.
A. Theta(n^2 log n)
B. Theta (n^2)
C. Theta(n^2 Log^2 n
D. Theta(n log^2 n)
Correct Answer: A. Theta(n^2 log n)

Q8: Programming, Data Structure and Algorithms (12)
Minimum (Asymptotic) time required to check whether a given binary tree is a binary search tree or not.
A. O(1)
B. O(n)
C. O(log n)
D. O(n log n)
Correct Answer: B. O(n)

Q9: Programming, Data Structure and Algorithms (13)
If you keep track of front, rear and count(no. of items in array), in a circular array when front =0 and rear=maxsize-1, what will be the count value.
A. 0

B. 3
C. 4
D. 2
Correct Answer C. 4

Q10: Programming, Data Structure and Algorithms (14)
Which of the following data structure can be applied directly on scheduling and resource allocation problems?
A. stack
B. queue
C. tree
D. linked list
Correct Answer: B. queue

Q11: Programming, Data Structure and Algorithms (15)
If you are asked to implement stack using queue how many queues are required to represent stack efficiently.
A. 1
B. 3
C. 4
D. 2
Correct Answer: D. 2

Q12: Programming, Data Structure and Algorithms (16)
Which is something you would NOT use a stack for?
i. Printing a name in reverse order
ii. A function call within another function call
iii. Arranging the elements of an array in increasing or decreasing order.
iv. Printing the elements of an array in the order they appear within the array.
A. only (i)
B. only (ii) and (iii) are correct
C. only (iii) and (iv) are correct
D. None of the options are correct
Correct Answer: C. only (iii) and (iv) are correct

Q13: Programming, Data Structure and Algorithms (17)
Which of the following type of queue is useful in building online multiplayer games.
A. Simple queue
B. Circular Queue
C. Priority Queue
D. Single Ended Queue

Correct Answer: B. Circular Queue

Q14: Programming, Data Structure and Algorithms (18)
In a linked list, there is a pointer to the first node, each node has pointer to the next node and last node has pointer to the first node, the name of the linkedlist is,
A. Singly linkedlist
B. Doubly linkedlist
C. Singly Circular linkedlist
D. Doubly Circular linkedlist
Correct Answer: C. Singly Circular linkedlist

Q15: Programming, Data Structure and Algorithms (19)
An array A of integers is given as A: array [1 … 5] [1 … 10]; The array is stored in column-major order and the first element of .The array is stored at location 50, Calculate the address of the element A[i][j] where each integer takes one memory location.
A. 5i + j + 44
B. i + 5j + 44
C. 5i + j + 39
D. 5j + i + 39
Correct Answer: B. i + 5j + 44

Q16: Programming, Data Structure and Algorithms (2)
In a binary tree, which of the following in terms of nodes, is not a valid argument?
A. It should have at least one node.
B. 2
C. 3
D. 0
Correct Answer: A. It should have at least one node.

Q17: Programming, Data Structure and Algorithms (20)
We are given a stack on which two operations are possible, Push( ) and Pop( ). Further, we are given an input sequence 7, 5, 2, 1, 3, 6, where 7 is the first element and 6 is the last element. Which sequence of elements cannot be printed using the given stack?
A. 6 3 1 2 5 7
B. 2 1 3 6 7 5
C. 5 2 7 1 3 6
D. None of the options
Correct Answer: B. 2 1 3 6 7 5

Q18: Programming, Data Structure and Algorithms (21)

Which of the given options provides the decreasing order of asymptotic complexity of functions f1, f2, f3 and f4?
f1=O(n),f2=O(log n), f3=O(n^2) f4=O(2n)
A. f1, f2, f3, f4
B. f3, f1, f4,f2
C. f1, f2, f4, f3
D. f1, f3, f2, f4
Correct Answer: B. f3, f1, f4,f2

Q19: Programming, Data Structure and Algorithms (22)
Which of the following methods have time complexity O(n) for a sorted array in descending order, where elements are distinct?
(i) Finding maximum element in the array.
(ii) Finding minimum element in the array.
(iii) Finding the mode of the array.
(iv) Find any element in an array.

```
Algorithm XY()
{
   i=1, j=1
  while(i<=n)
  {
    while(j<=n)
    {
     print("Chennai")
     j=j+1
    }
  }
}
```

A. (i)
B. (ii),(iii),(iv)
C. (iii),(iv)
D. (i),(iii)
Correct Answer: C. (iii),(iv)

Q20: Programming, Data Structure and Algorithms (23)
What is the infix version of the following postfix expression?
x 12 + z 17 y + 42 * / +
A. x + 12 + z / ((17 + y) * 42)
B. x + 12 + z / 17 + y * 42
C. x + 12 + z / (17 + y) * 42
D. x + (12 + z) / (17 + y * 42)
Correct Answer: A. x + 12 + z / ((17 + y) * 42)

Q21: Programming, Data Structure and Algorithms (24)
Suppose T1(n) = O(F (n)) and T2(n) = O(F (n)). Which of the following are true?
A. T1(n) + T2(n) = O(F (n))
B. T1(n) *T2(n) = O(F (n))
C. T1(n)=T2(n) = O(1)
D. T1(n) = O(T2(n))
Correct Answer: A. T1(n) + T2(n) = O(F (n))

Q22: Programming, Data Structure and Algorithms (25)
In a connected graph with no loops or multiple edges, which of the following inequalities is not correct? (v is the number of vertices, e is the number of edges)
A. $e \leq v \text{ ^2}$
B. $e \geq v - 1$
C. $v \geq e/2$
D. All the options
Correct Answer: C. $v \geq e/2$

Q23: Programming, Data Structure and Algorithms (26)
A circular doubly linked list has exactly two nodes, each of whose addresses are stored in two pointer variables, say head and tail. Which among the following statements is NOT True?
A. Head->next=tail
B. Head->prev->next=tail
C. Tail->prev->next=tail
D. Tail->next->prev=tail
Correct Answer: B. Head->prev->next=tail

Q24: Programming, Data Structure and Algorithms (27)
We have two stacks, named as A and B. Function Push(&A, data) pushes data on stack A, whereas function Push(&B, data) pushes data on stack B. Pop function on stack A pop out the element from stack A, and pushes it on stack B. On the other hand, Pop function on stack B pops out the element from stack B, and prints it on the screen. Which sequence of character will be printed on the screen after following set of operations: Push(&A, 8), Push(&B, 9), Push(&A, 4), Push(& B, 7), Push(&A,3), Pop(&A), Push(&B, 5), Pop(&A), Pop(&A), Pop(&B), Pop(&B), Pop(&B), Pop(&B), Pop(&B)?
A. 9 7 3 5 4 8
B. 8 4 5 3 7 9
C. 3 4 8 5 7 9
D. None of the options

Correct Answer: B. 8 4 5 3 7 9

Q25: Programming, Data Structure and Algorithms (28)
Which of the following will occur after the following set of operations on an empty stack of size 4. Push(2), Push (3), Pop( ), Push (3), Pop ( ), Pop( ), Push(2), Pop ( ), Pop ( ) ?
A. Stack underflow
B. Stack overflow
C. Operations will not lead to any error.
D. None of the options
Correct Answer: A. Stack underflow

Q26: Programming, Data Structure and Algorithms (29)
For the two functions f(n)=n^2+3n+4 and g(n)=n^3 we can conclude that,
A. f(n) = O(g(n))
B. f(n) = Θ(g(n))
C. g(n)=O(f(n))
D. g(n)= Θ(f(n))
Correct Answer: A. f(n) = O(g(n))

Q27: Programming, Data Structure and Algorithms (3)
In a stack 'S' the elements 4, 8, 12, 16 are pushed in the same order. Two times pop operation is executed, sum of the returned values is pushed again in stack S. Again two times pop operation is executed and sum of the returned values are pushed in stack S. The current top of the stack S is
A. 4
B. 16
C. 36
D. 28
Correct Answer: C. 36

Q28: Programming, Data Structure and Algorithms (30)
How many edges are there in a complete graph with 'n' vertices?
A. n*(n-1)/2
B. n*(n+1)/2
C. 2n
D. n/2
Correct Answer: A. n*(n-1)/2

Q29: Programming, Data Structure and Algorithms (31)

What is the value of the sum of the minimum in-degree and maximum out-degree of an Directed Acyclic Graph?
A. Will always be zero
B. Depends on a graph
C. Will always be greater than zero
D. None of the options
Correct Answer: B. Depends on a graph

Q30: Programming, Data Structure and Algorithms (32)
Traversal of a graph is different than tree because
A. There can be a loop in the graph
B. DFS on a graph uses stack, while inorder traversal is recursive
C. Both the given properties
D. None of the options
Correct Answer: A. There can be a loop in the graph

Q31: Programming, Data Structure and Algorithms (33)
Why is graph traversal generally more complex and difficult to implement than tree traversal?

A. Graphs can contain cycles and disconnected components, requiring algorithms to track visited nodes to avoid infinite loops.
B. Graphs lack a single, mandatory root node, meaning there is no predefined universal starting point for traversal.
C. Graphs can have multiple paths between the same two nodes, increasing the complexity of finding the most efficient route.
D. All of the above
Correct Answer: D All of the above

Q32: Programming, Data Structure and Algorithms (34)
The worst case time complexity of insertion sort is O(n^2). What will be the worst case time complexity of insertion sort if the correct position for inserting element is calculated using binary search?
A. O(n log n)
B. O(n)
C. O(n^2)
D. O(1)
Correct Answer: C. O(n^2)

Q33: Programming, Data Structure and Algorithms (35)
Consider the code given below:

```
void insertionsort(int arr[],int size)
{
int I,j,value;
```

```
for(i=1;i<size;i++)
{
value=arr[i];
j=i;
while(…)
{
arr[j]=arr[j-1];
j=j-1;
}
arr[j]=value;
}
}
```

what is line of code required for while statement to run the insertion sort

A. (j>0)||(arr[j-1]>value)
B. (j>0)&&(arr[j-1]>value)
C. (j>0)&&(arr[j+1]>value)
D. (j>0)&&(arr[j+1]<value)
Correct Answer: B. (j>0)&&(arr[j-1]>value)

Q34: Programming, Data Structure and Algorithms (36)
Consider an array of length 5, arr[5] = {9, 7, 4, 2, 1}. What are the steps of insertions done while running insertion sort on the array?

A. i)7 9 4 2 1
ii)4 7 9 2 1
iii) 2 4 7 9 1
iv) 1 2 4 7 9
B. i) 9 7 4 1 2
ii) 9 7 1 2 4
iii) 9 1 2 4 7
iv) 1 2 4 7 9
C. i) 7 4 2 1 9
ii) 4 2 1 9 7
iii) 2 1 9 7 4
iv) 1 9 7 4 2
D. i) 7 9 4 2 1
ii) 2 4 7 9 1
iii) 4 7 9 2 1
iv) 1 2 4 7 9
Correct Answer: A. i)7 9 4 2 1
ii)4 7 9 2 1
iii) 2 4 7 9 1
iv) 1 2 4 7 9

Q35: Programming, Data Structure and Algorithms (37)

A connected planar graph having 6 vertices, 7 edges contains _____ regions.
A. 11
B. 15
C. 1
D. 3
Correct Answer: D. 3

Q36: Programming, Data Structure and Algorithms (38)
When the entries 7, 4, 6, 1, 2, 3, 8, 5 are successively inserted into an initially empty binary search tree, what is the height of the resulting tree?
A. 4
B. 3
C. 7
D. 2
Correct Answer: A. 4

Q37: Programming, Data Structure and Algorithms (39)
How many different trees are possible with 10 nodes?
A. 512
B. 1024
C. 1014
D. 90
Correct Answer: C. 1014

Q38: Programming, Data Structure and Algorithms (4)
To implement Stacks using Queues, what are the minimum number of queues required?
A. 2
B. 3
C. 1
D. Atleast 3
Correct Answer: C. 1

Q39: Programming, Data Structure and Algorithms (40)
There are 8, 15, 13, 14 nodes in 4 different trees. Which of them could have formed a full binary tree?
A. 8
B. 15
C. 13
D. 14
Correct Answer: B. 15

Q40: Programming, Data Structure and Algorithms (41)
The memory use of an adjacency matrix is
A. O(n)
B. O(n^2)
C. O(n^3)
D. O(logn)
Correct Answer: B. O(n^2)

Q41: Programming, Data Structure and Algorithms (42)
A polynomial can be represented in a _____ by just storing the coefficient and exponent of each term.
A. Array
B. Linked list
C. Queue
D. Stack
Correct Answer: B. Linked list

Q42: Programming, Data Structure and Algorithms (43)
_____ is used to convert an infix expression into a postfix expression.
A. Linked list
B. Stack
C. Queue
D. Tree
Correct Answer: B. Stack

Q43: Programming, Data Structure and Algorithms (44)
Which type of linked list does not store NULL in next field?
A. Singly linked list
B. Circular linked list
C. Doubly linked list
D. Linear List
Correct Answer: B. Circular linked list

Q44: Programming, Data Structure and Algorithms (45)
The circular queue will be full only when
A. FRONT = MAX –1 and REAR = MAX –1
B. FRONT = 0 and REAR = MAX –1
C. FRONT = MAX –1 and REAR = 0
D. FRONT = 0 and REAR = 0
Correct Answer: B. FRONT = 0 and REAR = MAX –1

Q45: Programming, Data Structure and Algorithms (46)
Which type of linked list contains a pointer to the next as well as the previous node in the sequence?
A. Singly linked list
B. Circular linked list
C. Doubly linked list
D. All the options
Correct Answer: C. Doubly linked list

Q46: Programming, Data Structure and Algorithms (47)
In _____, insertions can be done only at one end, while deletions can be done from both the ends.
A. Double-ended queue
B. Input restricted deque
C. Output restricted deque
D. Circular queue
Correct Answer: B. Input restricted deque

Q47: Programming, Data Structure and Algorithms (48)
If the values of A, B, C, and D are 2, 3, 4, and 5, respectively, what is the value of the following prefix expressions: a. + – * A B C D
A. 7
B. -3
C. 5
D. 3
Correct Answer: A. 7

Q48: Programming, Data Structure and Algorithms (49)
What does the following lines of C code does?
CURSOR_SPACE[p].next = CURSOR_SPACE[0].next;
CURSOR_SPACE[0].next = p;
A. Allocates cursor space
B. Releases cursor space
C. Initialization of cursor space
D. Declaration for cursor
Correct Answer: B. Releases cursor space

Q49: Programming, Data Structure and Algorithms (5)
If you consider the following as number of operations that different algorithms take to complete a particular task on n elements, which one is linear ?
A. 3n^3+3n + 7

B. 3n^2 + 3n + 2
C. 3n + 1
D. 8
Correct Answer: C. 3n + 1

Q50: Programming, Data Structure and Algorithms (50)
Consider the linear arrays AAA(5:50) and BBB(-5:10). The number of elements in each array
A. 50, 18
B. 46, 18
C. 46, 16
D. 50, 16
Correct Answer: C. 46, 16

Q51: Programming, Data Structure and Algorithms (51)
Suppose the contents of an array A are, A = {1, null, null, null, null, 10}. What would be the size of the array considering it as a normal array and a sparse array?
A. 6 and 6
B. 6 and 2
C. 2 and 6
D. 2 and 2
Correct Answer: B. 6 and 2

Q52: Programming, Data Structure and Algorithms (52)
Consider the following code.

```
int fun()
{
  if(isEmpty())
  {
    return -10;
  }
 else
 {
   int n;
   n= q[front];
   front++;
   return n;
 }
}
```

Which operation does the above code perform?
A. Enqueue
B. Dequeue
C. Return the front element

D. Dequeue and return the front element
Correct Answer: D. Dequeue and return the front element

Q53: Programming, Data Structure and Algorithms (53)
In the Deque implementation using singly linked list, what would be the time complexity of deleting an element from the rear end?
A. O(1)
B. O(n^2)
C. O(n)
D. O(nlogn)
Correct Answer: C. O(n)

Q54: Programming, Data Structure and Algorithms (54)
In a circular queue implementation using an array of size 5, the array index starts with 0 where front and rear values are 3 and 4 respectively. Determine the array index at which the insertion of the next element will take place.
A. 5
B. 0
C. 1
D. 2
Correct Answer: B. 0

Q55: Programming, Data Structure and Algorithms (55)
Which of the following concepts make extensive use of arrays?
A. Binary trees
B. Scheduling of processes
C. Caching
D. Spatial locality
Correct Answer: D. Spatial locality

Q56: Programming, Data Structure and Algorithms (56)
Suppose that an intermixed sequence of push and pop operations are performed. The pushes push the integers 0 through 9 in order; the pops print out the return value. Which of the following sequences could not occur?
A. 4 3 2 1 0 9 8 7 6 5
B. 4 6 8 7 5 3 2 9 0 1
C. 2 5 6 7 4 8 9 3 1 0
D. 4 3 2 1 0 5 6 7 8 9
Correct Answer: B. 4 6 8 7 5 3 2 9 0 1

Q57: Programming, Data Structure and Algorithms (57)

Whenever there is a pending operation to be performed, the function becomes _____ recursive.
A. Direct recursive
B. Indirect recursive
C. Tail recursive
D. Non-tail recursive
Correct Answer: D. Non-tail recursive

Q58: Programming, Data Structure and Algorithms (58)
A node is said to be _____ if it has a possibility of reaching a complete solution.
A. Non-promising
B. Promising
C. Succeeding
D. Preceding
Correct Answer: B. Promising

Q59: Programming, Data Structure and Algorithms (59)
Which of the following is false in the case of a spanning tree of a graph G?
A. It is tree that spans G
B. It is a subgraph of the G
C. It includes every vertex of the G
D. It can be either cyclic or acyclic
Correct Answer: D. It can be either cyclic or acyclic

Q60: Programming, Data Structure and Algorithms (6)
In a circular queue with the capacity of n elements, insertion and deletion are performed using rear and front respectively. Initially rear and front are set to 1. Which of the following condition must be true for the queue to be full.
A. front=rear
B. front=rear+1
C. front=rear+1
D. rear=front+1
Correct Answer: B. front=rear+1

Q61: Programming, Data Structure and Algorithms (60)
Consider a complete graph G with 4 vertices. The graph G has ____ spanning trees
A. 15
B. 8
C. 16
D. 13
Correct Answer: C. 16

Q62: Programming, Data Structure and Algorithms (61)
A person wants to visit some places. He starts from a vertex and then wants to visit every place connected to this vertex and so on. What algorithm he should use?
A. Depth First Search
B. Breadth First Search
C. Prim's algorithm
D. Kruskal's algorithm
Correct Answer: B. Breadth First Search

Q63: Programming, Data Structure and Algorithms (62)
Which of the following is not an application of Depth First Search?
A. For generating topological sort of a graph
B. For generating Strongly Connected Components of a directed graph
C. Detecting cycles in the graph
D. Peer to Peer Networks
Correct Answer: D. Peer to Peer Networks

Q64: Programming, Data Structure and Algorithms (63)
Dijkstra's Algorithm cannot be applied on _____
A. Directed and weighted graphs
B. Graphs having negative weight function
C. Unweighted graphs
D. Undirected and unweighted graphs
Correct Answer: B. Graphs having negative weight function

Q65: Programming, Data Structure and Algorithms (64)
Consider the following code snippet
```
 char c[]="VITEEE2022";
 char *p=c;
 printf("%s",p+p[1]-p[3]);
```
A. EE2022
B. EEE2022
C. E2022
D. Error
Correct Answer: A. EE2022

Q66: Programming, Data Structure and Algorithms (65)
```
int main()
 {
        int s[5];
```

```
        //considering the initial memory location of s is 1000 and one integer is of 4 byte
        s++;
        printf("the array is at memory location %d",s);
        return 0;
}
```

Predict the output

A. 1002
B. 1004
C. 1020
D. Error

Correct Answer: D. Error

Q67: Programming, Data Structure and Algorithms (66)

```
int main()
{
        int s[5];
        //considering the initial memory location of s is 1000 and one integer is of 4 byte
        printf("%d %d",s+1,&s+1);
        return 0;
}
```

Predict the output

A. 1004 1020
B. 1004 1004
C. 1020 garbage value
D. Not even pass compilation

Correct Answer: A. 1004 1020

Q68: Programming, Data Structure and Algorithms (67)

```
#include
 int main ()
 {
        int arrayVal[]={5,8,12,14,18,19};
        int *pointerVal=(int*) (&arrayVal +1);
        printf("%d",*( pointerVal -1));
        return 0;
}
```

Output of the above will be

A. 12
B. 5
C. 19
D. 18

Correct Answer: C. 19

Q69: Programming, Data Structure and Algorithms (68)
What will be the output of the following code snippet

```
int main()
{
        int i;
        int s[5]={1};
        for(i=0;i<5;i++)
        {
                printf("%d",arr[i]);
        }
        return 0;
}
```

A. 1 followed by four garbage value
B. 1 0 0 0 0
C. 1 1 1 1 1
D. 0 0 0 0 0
Correct Answer: B. 1 0 0 0 0

Q70: Programming, Data Structure and Algorithms (69)
Consider the following statement

```
int (*p)[5];
```

What the above statement represents?
A. Compilation error
B. p is a pointer to 5 integers
C. p is an array of 5 pointers to integers
D. p is a pointer to an array of 5 integer
Correct Answer: D. p is a pointer to an array of 5 integer

Q71: Programming, Data Structure and Algorithms (7)
Consider the following pseudocode for the queue Q

```
Function(Q)
If (Q!=empty)
        x=delete(Q)
        Function(Q)
        Insert(Q,x)
```

What will be the change in a queue ?
A. Q element order is reversed
B. Q is unchanged
C. Q's Front element is removed and inserted at the rear, other elements are unchanged.
D. Q is emptied
Correct Answer: A. Q element order is reversed

Q72: Programming, Data Structure and Algorithms (70)

What will be the output of the following code snippet

```
#inlcude
void test(int *ptr)
{
        ptr=(int*)malloc(sizeof(int));
}
int main()
{
        int *ptr;
        test(ptr);
        *ptr=6;
        printf("%d n",*ptr);
        return(0);
}
```

A. May not work
B. Works and 6 will be printed
Correct Answer: A. May not work

Q73: Programming, Data Structure and Algorithms (71)
Consider the following code snippet and identify the true statement among the given options

```
int i;
int main()
{
        int j;
        int *k=(int *)malloc (sizeof(int));
}
```

A. i,j and *k are stored in stack segment
B. i and j are stored in stack segment. *k is stored on heap
C. i is stored in static area, j is stored in stack area and *k is stored in heap area.
D. j is stored in static area, i is stored in stack area and *k is stored in heap area.
Correct Answer: C. i is stored in static area, j is stored in stack area and *k is stored in heap area.

Q74: Programming, Data Structure and Algorithms (72)
Consider the following code snippet

```
#include
Int main()
{
        int num=5;
        int const * p=#
        ++(*p);
        printf("%d",num);
```

```
        return 0;
}
```
A. Compile time error
B. Run time error
C. 6
D. 5
Correct Answer: A. Compile time error

Q75: Programming, Data Structure and Algorithms (73)
Consider the following code snippet

```
#include
int main()
{
        int num=5;
        static int num1=num;
        if(num== num1)
                printf("equal");
        if(num> num1)
                printf("greater");
        else
                printf("less");
        return 0;
}
```
A. Compile time error
B. Equal
C. Greater
D. Less
Correct Answer: A. Compile time error

Q76: Programming, Data Structure and Algorithms (74)
Consider the following code snippet

```
#include
int test(int val)
{
        static int number=1;
        if(val >=5)
                return val;
        val = val +number;
        number++;
        return test(val);
}
```
What will be the value of test(1) will be:
A. 5
B. 6

C. 7
D. 8
Correct Answer: C. 7

Q77: Programming, Data Structure and Algorithms (75)
Consider the following code snippet

```
#include
int main(void)
{
        int num=10;
        const int * p=#
        *p=100;
        printf("i=%d",num);
        return 0;
}
```

A. i=100
B. i=10
C. compile time error
D. Runtime error
Correct Answer: C. compile time error

Q78: Programming, Data Structure and Algorithms (76)

```
Part1 (int i) {
      if (i == 0) return;
      printf ("%d", i);
      part2 (i - 2);
      printf ("%d", i);
}
Part2 (int i) {
      if (i == 0) return;
      printf ("%d", i;
      part1 (++i);
      printf ("%d", i);
}
```

What will be the output if part1(5) is called
A. 53423122233445
B. 53423120112233
C. 53423122132435
D. 53423120213243
Correct Answer: A. 53423122233445

Q79: Programming, Data Structure and Algorithms (77)

Consider the following code snippet

```
 int part(int n)
 {
    int x = 1, k;
    if (n = = 1) return x;
    for (k =1; k<n; ++k)
        x = x + fun(k) * fun(n–k);
    return x;
 }
```

What would be the output for fun(5)

A. 49
B. 50
C. 51
D. 52

Correct Answer: C. 51

Q80: Programming, Data Structure and Algorithms (78)

Consider the following code snippet

```
 int part(int x)
 {
    int i = 1, n;
    if (x = = 1) return i;
    for (n =1; n<x; ++n)
        i = i + fun(n) * fun(x–n);
    return i;
 }
```

What would be the output for fun(5)

A. 49
B. 50
C. 51
D. 52

Correct Answer: C. 51

Q81: Programming, Data Structure and Algorithms (79)

What will be the output of the following code?

```
#include<stdio.h>
#include<string.h>
 int main()
 {
      char str1[] = "VIT", str2[] = " Chennai ";
      //concatenates str1 and str2 and resultant string is stored in
str1.
      strcat(str1,str2);
      puts(str1);
      puts(str2);
      return 0;
 }
```

A. VIT Chennai Chennai
B. VIT
C. VIT Chennai
D. Compilation Error
Correct Answer: A. VIT Chennai Chennai

Q82: Programming, Data Structure and Algorithms (8)
Which of the following statement is more suitable while inserting in a circular queue
A. rear= rear +1
B. rear= rear-1
C. rear= (rear+1) mod SIZE
D. rear=rear mod SIZE +1
Correct Answer: C. rear= (rear+1) mod SIZE

Q83: Programming, Data Structure and Algorithms (80)
What will be the output of the C Program?

```
#include
int main()
{
        char arr[]=”function\0”;
        int num=strlen(arr);
        printf(“Length of function is %d”,num);
        return 0;
}
```

A. 9
B. 7
C. 10
D. 8
Correct Answer: D. 8

Q84: Programming, Data Structure and Algorithms (81)
Consider the following 2-D array initialization. Which of the options is correct?

```
int abc[2][] = {1, 2, 3 ,4 };
```

A. Valid Initalization
B. Invalid Initialization
Correct Answer: B. Invalid Initialization

Q85: Programming, Data Structure and Algorithms (82)
Consider the following 2-D character string

```
#define NUMBER_OF_STRING 5
#define MAX_STRING_SIZE 50
char arr[NUMBER_OF_STRING][MAX_STRING_SIZE] =
```

```
 {"Getting into the Data Structure", "is interesting ", "that make sure to effectively",
"organize the data and"“access & apply the operation”
 };
 for (int i = 0; i < NUMBER_OF_STRING; i++)
 {
 printf("'%s' has length %d\n", arr[i], strlen(arr[i]));
 }
```

What the above code will print

A. It will print the length of the string

B. It will print all strings in a string array with their lengths

C. It will print all the string in a string array

Correct Answer: B. It will print all strings in a string array with their lengths

Q86: Programming, Data Structure and Algorithms (83)

Structure in C language is

A. a collection of elements that can be of same data type

B. a collection of elements that can be of different data type

C. Elements of a structure are called members

D. All the options

Correct Answer: D. All the options

Q87: Programming, Data Structure and Algorithms (84)

The function of strncat and strncmp in string.h library is

A. Appends not more than n characters from the string pointed to by source to the end of the string pointed to by destination plus a terminating Null-character and Comparison is limited to the number of characters specified during the function call respectively.

B. Comparison is limited to the number of characters specified during the function call and appends not more than n characters from the string pointed to by source to the end of the string pointed to by destination plus a terminating Null-character respectively.

C. None of the options

Correct Answer: A. Appends not more than n characters from the string pointed to by source to the end of the string pointed to by destination plus a terminating Null-character and Comparison is limited to the number of characters specified during the function call respectively.

Q88: Programming, Data Structure and Algorithms (85)

Consider the following statements about array and structure and choose the correct statements

A. Array is a type of data structure used as container which can hold variables of same type whereas Structure data structure used as container which can hold variables of different types

B. Array it internally implements Pointer which always points to the first element of Array.

C. Structure is a user defined datatype
D. All the statements are true
Correct Answer: D. All the statements are true

Q89: Programming, Data Structure and Algorithms (86)
Which of the following is a properly defined struct?
A. struct {int a;}
B. struct a_struct {int a;}
C. struct a_struct int a;
D. struct a_struct {int a;};
Correct Answer: D. struct a_struct {int a;};

Q90: Programming, Data Structure and Algorithms (87)
What is the output of C program with structures.?

```
int main()
{
structure hotel
{
        int items;
        char name[10];
}a;
strcpy(a.name, "TAJ");
a.items=10;
printf("%s", a.name);
return 0;
}
```

A. TAJ
B. Empty string
C. Compiler error
D. None of the options
Correct Answer: C. Compiler error

Q91: Programming, Data Structure and Algorithms (88)
What will be the output of the program in C?

```
#include
int main()
{
        ant arr[5], i=0;
        while(i<5)
                arr[i]=++i;
        for(i=0;i<5;i++)
                printf(“%d,”,arr[i]);
```

```
        return 0;
}
```
A. 1,2,3,4,5
B. Garbage value, 1, 2, 3, 4,
C. 0, 1, 2, 3, 4,
D. 2, 3, 4, 5, 6,
Correct Answer: B. Garbage value, 1, 2, 3, 4,

Q92: Programming, Data Structure and Algorithms (89)
What is the output of this program?
```
#include
void main()
{
        int arr[10];
        printf("%d %d",arr[-2],arr[11]);
}
```
A. 0 0
B. Garbage value 0
C. Garbage value Garbage value
D. Compilation Error
Correct Answer: C. Garbage value Garbage value

Q93: Programming, Data Structure and Algorithms (9)
Which of the following recurrence cannot be solved using a master method?
A. T (n) = 2T (n/4) + n ^0.51
B. T (n) = 64T (n/8) – n^2 log n
C. T (n) = √ 2T (n/2) + log n
D. T (n) = 16T (n/4) + n!
Correct Answer: B. T (n) = 64T (n/8) – n^2 log n

Q94: Programming, Data Structure and Algorithms (90)
Let x be an array. Which of the following operations is illegal? (I) ++x (II) x+1 (III) x++ (IV) x*2
A. I and II
B. I, II and III
C. I, III and IV
D. III and IV
Correct Answer: C. I, III and IV

Q95: Programming, Data Structure and Algorithms (91)
What is the right way to Initialize array?
A. int num[6] = { 2, 4, 12, 5, 45, 5 };

B. int n{} = { 2, 4, 12, 5, 45, 5 };
C. int n{6} = { 2, 4, 12 };
D. int n(6) = { 2, 4, 12, 5, 45, 5 };
Correct Answer: A. int num[6] = { 2, 4, 12, 5, 45, 5 };

Q96: Programming, Data Structure and Algorithms (92)
How many stacks are required to implement a queue
A. 1
B. 2
C. 3
D. 4
Correct Answer: B. 2

Q97: Programming, Data Structure and Algorithms (93)

```
#include<stdio.h>
#include<string.h>
void main()
 {
        char str[]="VIT\0Chennai";
        printf("%s",str);
 }
```

What will be the output of the above program
A. VIT Chennai
B. Chennai
C. VIT\0Chennai
D. VIT
Correct Answer: D. VIT

Q98: Programming, Data Structure and Algorithms (94)
What is the output of C program with string

```
#include<stdio.h>
#include<string.h>
int main()
 {
 char str[]={'V','I','T'};
 printf("%s",str);
 return 0;
 }
```

A. VIT
B. V
C. VIT\0
D. None of the options
Correct Answer: A. VIT

Q99: Programming, Data Structure and Algorithms (95)
What is the output of following C program

```
#include<stdio.h>
#include<string.h>
 int main()
 {
 char str1[]="VIT";
 char str2[20];
 str2=str1;
 printf("%s",str2);
 return 0;
 }
```

A. VIT
B. VIT\0
C. Compile Error
D. V
Correct Answer: C. Compile Error

Q100: Programming, Data Structure and Algorithms (96)
what will be the output of the program

```
#include<stdio.h>
#include<string.h>
void main()
 {
    char str1[20]="VIT";
    char str2[20]=" Chennai";
          printf("%s",strcpy(str2,strcat(str1,str2)));
 }
```

A. ChennaiVIT
B. Chennai
C. VIT Chennai
D. VIT
Correct Answer: C. VIT Chennai

Q101: Programming, Data Structure and Algorithms (97)

```
#include<stdio.h>
#include<string.h>
int main()
 {
     char str1[] = "VIT", str2[] = " Chennai";
     strcat(str1,str2);
     puts(str1);
     puts(str2);
     return 0;
 }
```

A. VIT Chennai Chennai
B. VIT

C. VIT Chennai
D. Compilation Error
Correct Answer: A. VIT Chennai Chennai

Q102: Programming, Data Structure and Algorithms (98)
Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?
A. 90 and 99
B. 90 and 94
C. 89 and 99
D. 89 and 94
Correct Answer: A. 90 and 99

Q103: Programming, Data Structure and Algorithms (99)
The number of swapping needed to sort the numbers 5,1,4,3,2 in ascending order, using bubble sort is:
A. 10
B. 9
C. 8
D. 7
Correct Answer: D. 7